



Numerical Methods · Worksheet 1

Chapter 9: Numerical Methods. Answer all questions. Show your working clearly. Total: 107 marks. Time: 2 hours 15 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are five sections, one for each topic in Chapter 9. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- When using a change of sign, always state that the function is **continuous** and give a conclusion.
- Write down each iterate to at least 4 decimal places. All angles are in radians.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Locating roots by a change of sign	18	
B Fixed-point iteration	21	
C Cobweb and staircase diagrams, and convergence	22	
D The Newton–Raphson method	22	
E When numerical methods fail	20	
Roar-level challenge	4	
Total	107	

Section A Locating roots by a change of sign

18 marks

1. Show that the equation $x^3 + x - 3 = 0$ has a root between $x = 1$ and $x = 2$.

[2]

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2. Show that the equation $e^x - 3x = 0$ has a root in the interval $[1.5, 1.6]$. [3]

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3. $f(x) = x^3 - 5x + 1$. The equation $f(x) = 0$ has a root α in the interval $[2.1, 2.2]$. [3]
By choosing a suitable interval, show that $\alpha = 2.13$ correct to 2 decimal places.

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4. Given that $\tan 1 > 0$ and $\tan 2 < 0$, a student concludes that the equation $\tan x = 0$ has a root between $x = 1$ and $x = 2$. [3]
Explain why the student is wrong.

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5. $f(x) = \ln(x+2) - x + 1$, $x > -2$.

(a) Show that the equation $f(x) = 0$ has a root α in the interval $[2, 3]$. [2]

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(b) Show that $\alpha = 2.51$ correct to 2 decimal places. [3]

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(c) Show that the equation $f(x) = 0$ has a second root in the interval $[-1.99, -1.9]$. [2]

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(Total for Question 5 is 7 marks)



Section B Fixed-point iteration

21 marks

6. Using the iteration formula $x_{n+1} = \sqrt{2x_n + 3}$ with $x_1 = 1$, find the values of x_2 , x_3 and x_4 . [2]

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7. (a) Show that the equation $x^3 - 4x^2 - 1 = 0$ can be written in the form $x = 4 + \frac{1}{x^2}$. [3]

- (b) Using $x_{n+1} = 4 + \frac{1}{x_n^2}$ with $x_0 = 4$, find the root correct to 4 decimal places.

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8. Show that the equation $x^2 - 5x + 2 = 0$ can be written as $x = \frac{x^2 + 2}{5}$. [3]

- Use the iteration $x_{n+1} = \frac{x_n^2 + 2}{5}$ with $x_0 = 0.5$ to find a root correct to 3 decimal places.

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9. Show that the equation $e^x + x - 4 = 0$ can be rearranged to give $x = \ln(4 - x)$.
Using $x_{n+1} = \ln(4 - x_n)$ with $x_0 = 1$, find the root correct to 3 decimal places.

[4]

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10. $f(x) = x^3 + 3x - 12$

- (a) Show that the equation $f(x) = 0$ has a root α in the interval $[1.8, 1.9]$. [2]

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- (b) Show that the equation $f(x) = 0$ can be written as $x = \sqrt[3]{12 - 3x}$. [2]

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- (c) Using the iteration formula $x_{n+1} = \sqrt[3]{12 - 3x_n}$ with $x_1 = 2$, find x_2 , x_3 and x_4 , giving your answers to 4 decimal places. [3]

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- (d) By choosing a suitable interval, show that $\alpha = 1.859$ correct to 3 decimal places. [2]

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(Total for Question 10 is 9 marks)

**Section C Cobweb and staircase diagrams, and convergence**

22 marks

11. An iteration $x_{n+1} = g(x_n)$ has a fixed point L with $g'(L) = -0.4$. [2]

State whether the iteration converges to L from a nearby starting value, and describe the shape of the resulting diagram.

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12. For the iteration $x_{n+1} = \frac{x_n^2 + 2}{5}$ from Question 8, the root is $L \approx 0.438$. [3]

Find $g'(L)$, and hence explain why the iteration converges, stating whether the diagram is a staircase or a cobweb.

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13. The equation $x^3 - 4x - 1 = 0$ has a root $\beta \approx 2.115$. [3]

Explain why the iteration $x_{n+1} = \frac{x_n^3 - 1}{4}$ will not converge to β .

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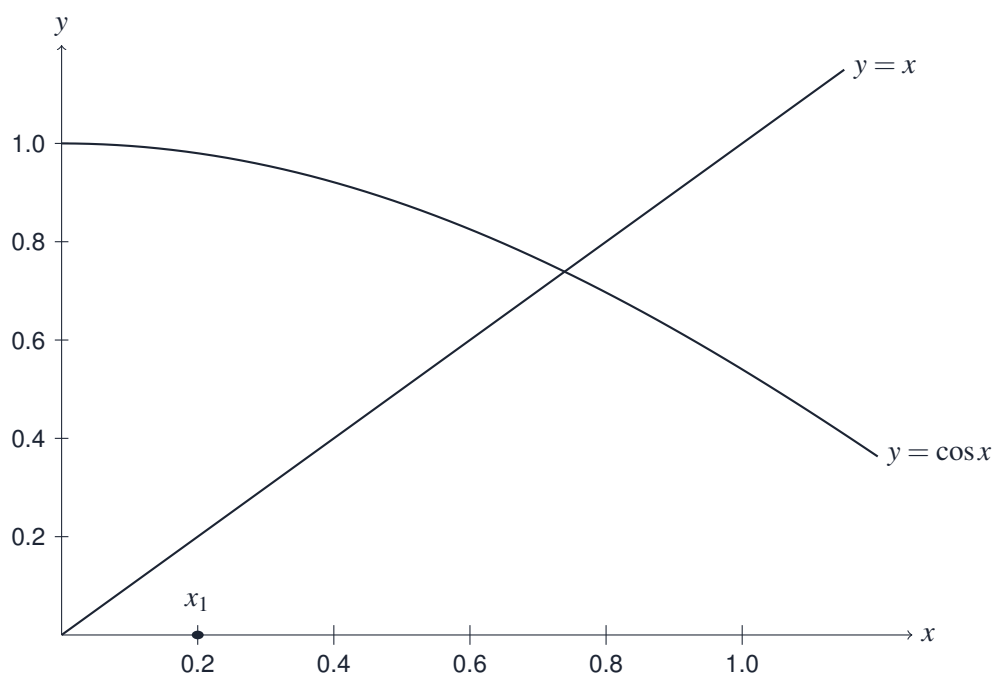
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14. The diagram shows parts of the graphs of $y = \cos x$ and $y = x$. The iteration $x_{n+1} = \cos x_n$ is used with $x_1 = 0.2$. [4]



- (a) On the diagram, draw a cobweb or staircase diagram to show how the iteration converges, and mark the positions of x_2 and x_3 on the x -axis.
- (b) The root is $L \approx 0.739$. Calculate $g'(L)$ and explain how its value is consistent with your diagram.

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15. The equation $x^3 - 3x - 5 = 0$ has a single real root α .

(a) Show that α lies between 2 and 3. [2]

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(b) Two possible rearrangements of the equation are [2]

A: $x = \frac{x^3 - 5}{3}$ B: $x = \sqrt[3]{3x + 5}$.

Show that both rearrangements are equivalent to the original equation.

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(c) By considering the gradient of each iteration function near $x = 2.3$, determine which rearrangement will give an iteration that converges to α . Justify your answer. [4]

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(d) Use the convergent iteration with $x_0 = 2$ to find α correct to 2 decimal places. [2]

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(Total for Question 15 is 10 marks)



Section D The Newton–Raphson method

22 marks

16. Use the Newton–Raphson method with $f(x) = x^2 - 7$ and $x_0 = 3$ to find x_1 and x_2 , and hence give an approximation for $\sqrt{7}$. [3]

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17. Use the Newton–Raphson method with $x_0 = 2$ to find the root of $x^3 - 2x - 5 = 0$ correct to 3 decimal places. [3]

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18. $f(x) = e^x - 3x - 1$. Using $x_0 = 1.5$, apply the Newton–Raphson method to find a root of $f(x) = 0$ correct to 4 decimal places. [4]

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19. Use the Newton–Raphson method with $x_0 = 2$ to find the positive root of $\sin x - \frac{x}{2} = 0$ correct to 4 decimal places. [4]

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20. $f(x) = \ln x + x^2 - 4$, $x > 0$.

(a) Show that the equation $f(x) = 0$ has a root α in the interval $[1.8, 1.9]$. [2]

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(b) Using $x_0 = 1.8$ as a first approximation, apply the Newton–Raphson method twice to find x_2 , [4]
giving your answer to 4 decimal places.

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(c) By choosing a suitable interval, show that your value of x_2 is correct to 4 decimal places. [2]

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(Total for Question 20 is 8 marks)



Section E When numerical methods fail

20 marks

21. Explain why the Newton–Raphson method cannot be applied to $f(x) = x^3 - 3x + 1$ with starting value $x_0 = -1$. [2]

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22. $f(x) = x^2 - 2.6x + 1.68$. A student finds that $f(1) > 0$ and $f(2) > 0$, and concludes that there is no root between $x = 1$ and $x = 2$. [3]

Show that the student is wrong, and explain why the change-of-sign method did not detect a root.

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23. The equation $x^2 - 5x + 6 = 0$ has roots $x = 2$ and $x = 3$. [3]

Using $x_{n+1} = \frac{x_n^2 + 6}{5}$ with $x_0 = 3.5$, find x_1 , x_2 and x_3 .

Explain, with reference to the gradient of the iteration function at $x = 3$, what you notice.

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24. $f(x) = x^3 - 6x^2 + 9x - 3$

[4]

- (a) Explain why $x_0 = 3$ cannot be used as a starting value for the Newton–Raphson method.
- (b) A student uses $x_0 = 2.9$. Calculate x_1 , and explain why this starting value is a poor choice for finding the root near $x = 3.9$.

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25. $f(x) = x^3 - 12x + 5$

- (a) Show that the equation $f(x) = 0$ has a root α in the interval $[3, 4]$.

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- (b) Using $x_0 = 3$, apply the Newton–Raphson method to find α correct to 3 decimal places.

[3]

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- (c) A student instead uses $x_0 = 2.1$. Find x_1 , and use the shape of the curve $y = f(x)$ near $x = 2$ to explain why this is an unsuitable starting value.

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(Total for Question 25 is 8 marks)



ROAR-LEVEL CHALLENGE

4 marks

26. Show that applying the Newton–Raphson method to $f(x) = x^2 - a$, where $a > 0$, gives the iteration [4]

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right).$$

Hence, using $x_0 = 2$, find $\sqrt{5}$ correct to 4 decimal places.

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END OF WORKSHEET

Total: 107 marks